

KAIRUI FENG

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EDUCATION

Zhejiang University

Expected Jun. 2027

B.Eng. in Information Security, College of Computer Science and Technology

GPA: 3.96/4.30 Weighted Average: 87.33/100 IELTS Academic: 6.5

SECURITY-RELATED CORE COURSEWORK

Security Coursework Average: 92.1/100 (18.5 credits)

Software Security Principles (99); Software Security Theory & Practice (92); Network Security Theory & Practice (92); AI Security (89); AI Security & Ethics (89); Advanced Cryptology (92); System Security (97); Wireless/IoT Security (84); Privacy, Security & Digital Currency (95).

SELECTED SYSTEMS PROJECTS AND RESEARCH EXPERIENCE

Linux Kernel Security Analysis and Exploitation

Independent Security Practice

- Analyzed kernel coloring techniques through source-code reading.

gdb-helper

Open-Source Debugging Tool

- Extended an open-source GDB plugin with DeepSeek API integration to translate natural-language CLI requests into multi-step native GDB workflows. <https://github.com/fkrcarry/gdb-helper>

RISC-V Operating System on a Self-Designed CPU
Curriculum

ZJU Computer Systems Integrated

- Independently built a RISC-V OS running on a self-designed CPU. **SYS1:** designed and validated a single-cycle RISC-V CPU in HDL.
- **SYS2:** implemented a basic kernel with interrupts and context switching; built an RV64I CPU with a five-stage pipeline, branch prediction, bus interface, forwarding, L1 cache, and privileged instructions.
- **SYS3 (Labs 1–5 and xpart MMU):** implemented boot and timer interrupts, preemptive kernel threads, virtual memory, user mode and system calls, page faults, process creation, and MMU functionality.
- Received the *Outstanding Project Award* from Prof. Rui Chang and Prof. Wenbo Shen (9 recipients among approximately 150 students).

C++ RISC-V Compiler

Independent Systems Project

- Developed a C++ compiler targeting RISC-V.

Mobile GUI-Agent-Based iOS App Behavior-Transformation Detection System
Prof. Haitao Xu

Under

- Developed a mobile GUI-agent-based detector for iOS apps that change behavior or presentation during interaction.

MCP Server Security Vulnerability Discovery

Independent Security Research

- Identified and publicly disclosed an SSRF vulnerability in `imprvhub/mcp-browser-agent` (up to 0.8.0) through user-controlled fields in `CallToolRequestSchema`; CVE-2026-5607: https://vulldb.com/source_cve/355398

Other Security Research

Zhejiang University

- Code LLM red-teaming under Prof. Kui Ren and Prof. Zhan Qin (patent pending); SEED Labs security experimentation under Prof. Wenliang Du.

TEACHING EXPERIENCE

Teaching Assistant, Zhejiang University: Computer Systems I (SYS1), Computer Systems II (SYS2), and Principles and Practice of Software Security.

Delivered theory and laboratory instruction; released and checked labs; graded coursework and quizzes.

HONORS AND AWARDS

- University-Level Student Pacesetter Award, Zhejiang University (twice)
- Second Prize, National College Student Information Security Competition
- First Prize, Zhejiang Provincial College Student Information Security Competition
- Third Place, Pwn Track, ZJUCTF 2025 (Zhejiang University)
- Ranked No. 4 in the Pwn track on Zhejiang University's CTF platform
- Outstanding Project Award, Computer Systems Integrated Curriculum

TECHNICAL SKILLS

Programming Languages C, C++, Python, SystemVerilog

Systems & Tools Linux command line, Git, Docker, $\text{\LaTeX}$, QEMU kernel debugging, binary debugging, kernel vulnerability reproduction/analysis, fuzzing, Linux kernel modules/drivers, kernel privilege-escalation exploitation